SvaraAI

A voice-first, multimodal dining-accessibility platform for Indian restaurants — built on Bhashini (ASR · NMT · TTS · OCR) with on-device Android intelligence

1. Problem and Our Understanding

Restaurant menus in India are physical, dense and multilingual, and the people who most need independence at a table are the ones least served by them. Four barriers combine to exclude diners from the full dining lifecycle (discover → ask → order → get service → pay):

Barrier	Who is affected	What goes wrong at the table
Visual impairment	~34 M severe visual impairment/blindness; ~40 M low vision	Laminated 8-pt multi-column menus cannot be read; total dependence on companions or rushed staff.
Ageing and hearing/cognitive decline	140 M+ senior citizens	Long menus tire working memory; fast speech is unintelligible; small flat buttons are hard to hit.
Language divide	Diners vs migrant kitchen/service staff (Telugu, Tamil, Kannada, Malayalam vs Hindi, Odia, Bengali)	At 75–85 dB ambient noise, diner and waiter cannot understand each other; deaf/mute diners have no channel at all.
Financial and allergen opacity	Everyone, but critically Jain, nut-allergic and vegan diners	GST (2.5% CGST + 2.5% SGST) and service charge are hard to audit; hidden cashew gravy or onion/garlic in “Jain” dishes causes real harm.

2. Our Idea

SvaraAI is a native Android app that lets a blind, elderly, deaf, mute or non-local diner complete an entire restaurant visit **independently and safely, in their own language**. The diner points a phone at a physical menu and talks; SvaraAI frames the shot with spoken coaching, reads the menu into a structured catalogue, answers dietary questions, takes the order, bridges the conversation with staff, and splits the bill. Three design decisions define it:

- **Bhashini-first language stack.** Government-of-India Bhashini models (AI4Bharat Conformer ASR, IndicTrans2 NMT, IITM FastSpeech2 TTS, Indic OCR) serve Telugu (default), Hindi, Tamil, Kannada, Malayalam and Indian English, wrapped by a restaurant-specific lexical normaliser that fixes the errors generic models make on food words.
- **Offline-safe by construction.** Every cloud dependency has a latency budget and an automatic on-device fallback, so a basement restaurant never produces silence.
- **Deterministic where it matters.** Money is integer-paise arithmetic, allergen answers come from a verified encyclopedia + guarded LLM, and the LLM is food-only with temperature 0.1.

3. System Architecture

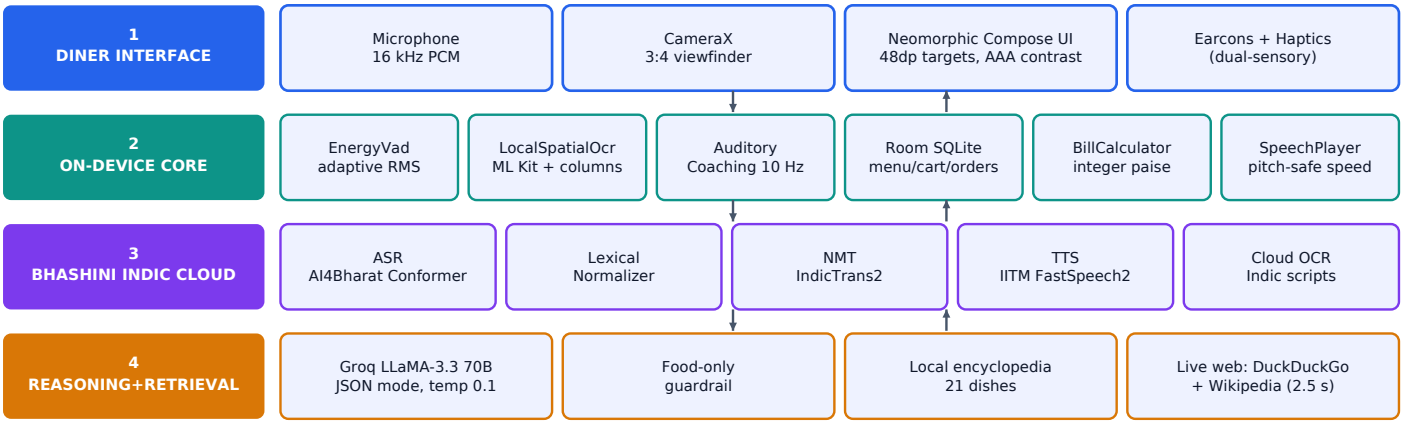


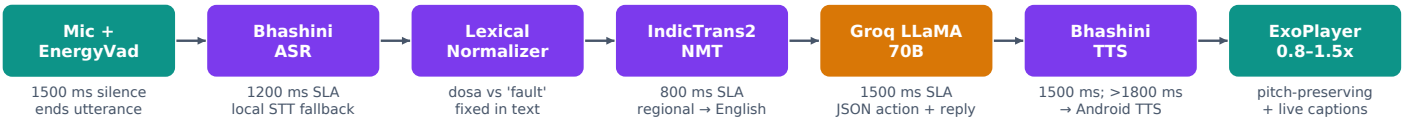
Fig. 1 — Four-layer clean architecture (Kotlin, Jetpack Compose, MVI unidirectional data flow, Room, DataStore). Down-arrows: input and request flow; up-arrows: spoken, visual and haptic response.

4. Core Features and How They Work

4.1 Voice assistant with noise-robust acoustic front-end

Raw 16 kHz mono PCM is cut into 20 ms frames (320 samples). **EnergyVad** computes RMS per frame, calibrates the room’s noise floor over the first 10 frames (200 ms), and sets an adaptive threshold `clamp(2.2 × noiseFloor, 350, 2000)`. Speech starts only after **3 consecutive** loud frames (rejects cutlery clinks) and ends after **1500 ms** of silence or 12 s total, so silence is detected on-device with no cloud streaming. Audio lives only in heap memory and is never written to disk. Wake word “Hey Svara” or a large tactile mic button starts a turn.

4.2 Bhashini Indic pipeline: one voice turn end to end



Earcons: 880 Hz start chime → 587 Hz stop tone → C-major triad on order confirmed → dissonant beat on error; haptics mirror each cue.

Fig. 2 — Voice turn with per-stage latency budgets (ASR: AI4Bharat Conformer, Whisper-medium for English; NMT: IndicTrans2; TTS: IITM FastSpeech2, 22.05 kHz). Service IDs and tokens come from ULCA getModelsPipeline, cached 24 h.

Restaurant Lexical Normaliser — our key language contribution. Generic ASR/NMT models are trained on parliamentary and news text, so casual code-switching breaks them: the Telugu word for dosa collides with *doshamu* (fault), so “two dosas please” is translated as *“two mistakes”*; *poori* becomes *“complete”* and *vada* becomes *“wind”*. `RestaurantLexicalNormalizer`

intercepts these tokens per language (te, hi, ta, kn) and forces the correct dish name (dosa, idli, poori, vada, chai, filter coffee), and smooths passive translations ("You have to have…") into direct diner intent ("I want…"). **NumberVerbalizer** speaks prices grammatically in each language (₹250 is spoken as "two hundred and fifty rupees" in each language's own number words, e.g. Hindi "do sau pachaas rupaye") instead of letter-by-letter English digits. Playback uses ExoPlayer's Sonic time-stretching with pitch pinned at 1.0, exposing **0.8x–1.5x** speed: slow for elders, fast for expert screen-reader users.

4.3 CameraX menu scanner, spatial OCR and blind-friendly coaching

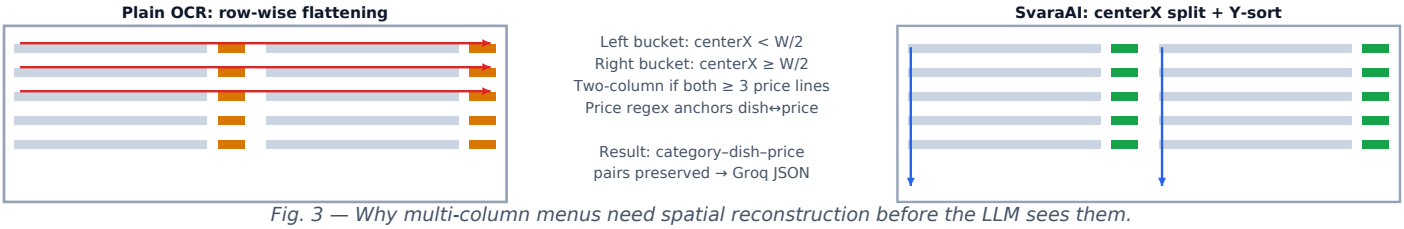


Fig. 3 — Why multi-column menus need spatial reconstruction before the LLM sees them.

- **Auditory coaching (10 Hz):** nine localised states (PERFECT_ALIGNMENT, TOO_DARK, TOO_BRIGHT, BLURRY, MOVE_UP/DOWN/LEFT/RIGHT, SEARCHING) spoken in Telugu/Hindi/English; a haptic TICK fires when all four corners are contained and contrast is stable, then capture is triggered.
- **Dual-engine OCR:** Google ML Kit (bounding boxes, on-device) and Bhashini cloud OCR (Indic scripts) run in parallel on a 1600 px, EXIF-corrected, contrast-stretched image. A price regex anchors prices; the page is split at W/2 and each column is Y-sorted so dish↔price pairs survive.
- **LLM structuring:** Groq `llama-3.3-70b-versatile` (JSON mode, temp 0.1) converts noisy text into sections and items with English/Hindi/Telugu names, price in paise, veg/Jain/vegan/nut-free flags, spice 0–3 and chef-special; stored transactionally in Room. A timed funnel then says "Ready to Order" (3 s) → "Compare Dishes", avoiding cognitive overload.

4.4 Smart dish comparison

Side-by-side cards compare price (absolute $\Delta = |P_A - P_B|$, relative $\Delta\% = \Delta / \max(P_A, P_B) \times 100$), dietary badges, spice-flame gauge (0–3), key preparation and nut-free status, with a spoken narration ("Masala Dosa is cheaper by one hundred and thirty rupees") and one-tap Add-to-Cart. A catalogue mode adds Flow-driven filters (VEG, NON_VEG, JAIN, NUT_FREE; MILD/MEDIUM/SPICY; price sort). An 8-dish sample menu keeps the feature usable before any scan.

4.5 Dietary and allergen safety guard (three-tier intelligence)

Tier	Source	Behaviour
0 · Guardrail	isFoodQuery check	Politics, coding, sports, maths etc. get a polite spoken redirection in the diner's language ("I am Svara, your restaurant assistant…").
1 · Cache	Local encyclopedia, 21 dishes (Chilli Chicken, Biryani, Dal Makhani, Rasam …)	Instant, offline, human-verified dietary class, spice, origin in te/hi/en.
2 · Live web	DuckDuckGo Instant Answer + Wikipedia REST, 2.5 s hard timeout	For unlisted dishes (Pesarattu, Bisi Bele Bath, Gongura Mutton); filler words stripped from the query first.
3 · Reasoning	Groq LLaMA 3.3 70B with strict dietary system prompt	Fuses retrieved text into a verified answer plus UI actions (HIGHLIGHT_ITEM, FILTER_JAIN/VEG/NUT_FREE, NAVIGATE_SECTION).

It encodes India-specific risks: hidden **cashew (kaju) gravies** in korma/butter chicken/shahi paneer, **Jain** exclusion of all root vegetables (onion, garlic, potato, ginger…), ghee/butter for vegans, and groundnut cross-contact. Where a dish is uncertain (e.g. Dal Makhani often has garlic), SvaraAI says so and prompts the diner to confirm special preparation with the waiter instead of guessing.

4.6 Unified Order and Service Hub (cart · waiter bridge · bill)

One screen with three tabs removes the disorienting screen-hopping of earlier designs; state is shared through Room flows.

- **Voice cart:** "Add two Masala Dosa with less oil and one Filter Coffee extra hot" → Groq extracts item, quantity and cooking note; the cart is stored as `CartItemEntity`; a narration button speaks the whole cart with subtotal, taxes and notes; the C-major earcon plus haptic confirms each change.
- **Waiter Bridge:** dual-mic bar. "Speak to Waiter" = ASR → IndicTrans2 → TTS in the staff language (e.g. Telugu → Hindi with honorifics such as "Bhaiya"); "Waiter Response" reverses the path with live captions. An outward card rotated 180° lets the phone lie flat facing the waiter, and one-tap phrase chips (cold food, delayed order, missing dish, too spicy, clean table, compliments, urgent manager call) work for mute or deaf diners.
- **Smart bill:** every amount is a Long in paise, so there is no floating-point drift (0.1 + 0.2 ≠ 0.3 in Double).

Quantity	Formula (integer paise)	Quantity	Formula (integer paise)
Subtotal	Σ unitPrice$_i$ × qty$_i$	**Service charge (optional 5%)**	⌊ Subtotal × 50 / 1000 ⌋
CGST 2.5%	⌊ Subtotal × 25 / 1000 ⌋	**Grand total**	Subtotal + CGST + SGST + Service
SGST 2.5%	⌊ Subtotal × 25 / 1000 ⌋	**Equal split of T among N**	Base = ⌊T/N⌋; first R = T mod N diners pay +1 paisa; Σ shares = T exactly

Itemised mode assigns dishes to people and allocates tax, service charge and tip in proportion to each person's subtotal; the result is read aloud as natural spoken rupees and can be paid by UPI request.

5. Accessibility Engineering

Concern	Design response
Low vision / tremor	Neomorphic dual-bevel physics (convex rest, concave pressed) with ≥ 48 × 48 dp targets (WCAG 2.5.5), text contrast 14.8:1 and 7.2:1, no emojis (screen readers mis-announce them), shape-and-label dietary badges (veg square, non-veg diamond, Jain badge) so colour is never the only cue.
Blind / deaf	Every state change has a synchronised earcon (44.1 kHz PCM synthesised in memory, zero asset I/O) and a haptic pattern (CLICK, CONFIRMATION, CAMERA_STABLE, ERROR_ALERT, SPEECH_TURN_CUE); live caption bar mirrors all speech.
Vestibular / cognitive	9-card onboarding pager with gentle scale/alpha transitions, one-question-at-a-time funnels, adjustable speech rate, native-script language selector at the top of every screen.

6. Reliability, Privacy and Security

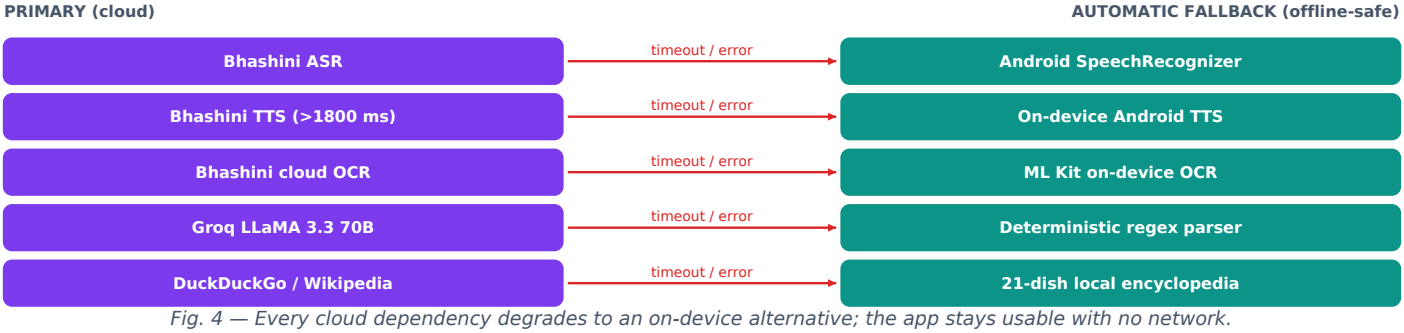


Fig. 4 — Every cloud dependency degrades to an on-device alternative; the app stays usable with no network.

Privacy: audio is held only in transient memory and wiped after transmission, never persisted or sent to analytics. **Keys:** Bhashini and Groq credentials are injected through BuildConfig and protected with Android KeyStore. **Safety:** the LLM is domain-locked to food and menu tasks, with temperature 0.1 and JSON-schema output.

7. Results and Verification

Test suite (JUnit, 15 suites)	Verifies	Test suite	Verifies
GroqClientDietaryTest (21)	Jain exclusion, cashew flags, cart actions	Phase7VoiceCartTest (6)	Speech-to-cart, quantities, notes
CulinaryQaAndGuardrailsTest (15)	Non-food redirect, Indic queries, allergens	Phase10WaiterBridgeTest (6)	Two-way translation, card flip, chips
Phase12DynamicAdaptationTest (15)	Elderly slow speech, contrast, sensory triggers	EnergyVadTest (5)	Adaptive RMS, 1500 ms timeout
WakeWord suites (9 + 8)	“Hey Svara”, multi-dialect phonetics, goodbye	Phase11BillPaymentTest (4)	Paise math, 2.5% + 2.5% GST
Phase4AdapterTest (7)	Dish transliteration protection, chunking	Onboarding (3), Coaching (3), OCR (3)	Pager, framing cues, column clustering
CulinaryWebSearchClientTest (7)	DDG, Wikipedia, 2.5 s timeout	Phase6NavigatorTest (2)	Category voice jumping

Outcomes. All unit-test suites pass (100+ tests) across voice, language, OCR, safety, ordering and billing. Enforced latency budgets: ASR 1.2 s, NMT 0.8 s, TTS 1.5 s (fallback at 1.8 s), LLM 1.5 s, web lookup 2.5 s. Money correctness is checked by the invariant Σ shares = total. The Telugu “dosa → mistake” failure is eliminated by the normaliser.

8. Field Scenarios and Impact

Persona	Journey with SvaraAI
Blind engineer, Hyderabad mess	Coach says “menu cut off on right, move phone right” → tick when framed → “4 sections, 28 items loaded” → “Add two idli and one filter coffee” → triad chime. No sighted help needed.
Telugu tourist, Hindi-speaking waiter	Speaks Telugu; phone announces natural Hindi with the honorific “Bhaiya”; waiter’s Hindi reply returns as Telugu captions and speech.
76-year-old grandmother, Bangalore	Speech at 0.8x with natural pitch, large tactile cards, high-contrast captions, no cognitive overload.
Strict Jain family, Mumbai	Asks about Dal Makhani and Paneer Butter Masala; app flags garlic and cashew paste and tells them to confirm special preparation.

Why it matters. SvaraAI turns India’s own public AI infrastructure (Bhashini) into dignity at the dining table: more than 100 million people with vision, hearing, age or language barriers gain independent discovery, safe dietary choice, verifiable billing and a shared language with staff. It needs no restaurant integration, because the physical menu is the interface. **Stack:** Kotlin, Jetpack Compose, CameraX, ML Kit, Room, DataStore, ExoPlayer, OkHttp, Bhashini ULCA/Dhruva, Groq. **Next steps:** more Indic languages (Bengali, Marathi, Odia), on-device quantised ASR and a restaurant-side order sync.

9. Key Innovations

Innovation	Why it is different
Restaurant lexical normaliser	Fixes the dosa/“fault”, poori/“complete”, vada/“wind” failures of general Bhashini ASR/NMT output, making Indic food conversation reliable.
Blind-first spatial OCR	A 10 Hz spoken coaching loop plus column-aware clustering lets a blind user capture and parse a multi-column laminated menu unaided.
Dual-sensory feedback	Synthesised earcons and Android haptic compositions give every state a sound and a touch cue, serving blind and deaf users equally.
Deterministic money and safety	Integer-paise GST with exact remainder allocation; food-only LLM guardrail; cashew and Jain checks backed by a verified encyclopedia.
Offline-safe degradation	Five cloud-to-device fallbacks and strict timeouts mean the assistant never freezes in a noisy or low-signal restaurant.

Reproducibility. JDK 17, Android Studio; `./gradlew assembleDebug` builds the APK and `./gradlew test` runs the full unit-test verification. Bhashini (ULCA user ID and API key, inference key) and Groq keys are supplied through build configuration; setup details are in the repository README.